1. Read a text file and count the frequency of each word using dictionaries.
2. Implement matrix multiplication using NumPy arrays.
3. Generate a random array and compute mean, median, variance, and standard deviation.
4. Create a 1D array of 25 elements and reshape it into a 5×5 matrix.
5. Demonstrate slicing operations on a 2D NumPy array.
6. Generate 100 random numbers from a normal distribution and plot their histogram.
7. Create two arrays and perform addition, subtraction, multiplication, and division element-wise.
8. Generate a 3×3 matrix and compute its transpose.
9. Compute the dot product of two vectors using NumPy.
10. Create a 5×5 identity matrix and verify its properties.
11. Extract the diagonal elements of a 2D NumPy array.
12. Plot a sine and cosine curve on the same graph.
13. Plot a bar chart showing marks of 5 students.
14. Plot a scatter plot of two features (e.g., height vs weight).
15. Plot the distribution of exam scores using a histogram.
16. Plot a pie chart showing percentage of expenses in different categories.
17. Plot a box plot to detect outliers in a dataset.
18. Generate 100 random points in 2D using NumPy and visualize them with a scatter plot.
19. Generate 100 random points in 3D and visualize them.
20. Generate a 10×10 random matrix and visualize with a heatmap.